

Chapter 9 In-Class Exercise: Course Schedules and Modules

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1 Adding Meeting Times

1. Extend the course records from the warm-up by adding the meeting days and meeting time:

```
[subject,number,building,room,days,time]
```

For example,

```
course = ["PHYS",100,"SLC",202,"MWF",1400]
```

represents Physics 100, meeting in SLC 202 on Monday, Wednesday, and Friday at 1400 (2:00 PM). Military time is useful because numerical times can be sorted directly:

```
930 < 1100 < 1400 < 1530
```

2 Build the Module

Create a new Python file called `course_tools.py`. All of our course-management functions will be placed inside this module. Begin with the function from the warm-up:

```
def add_course(courses, course):  
    courses.append(course)
```

Now define a function that returns the meeting time:

```
def get_time(course):  
    return course[5]
```

Use it to sort the master list:

```
def sort_by_time(courses):  
    courses.sort(key=get_time)
```

The key tells Python which part of each record should control the sorting.

3 Displaying the Schedule

Add one final function:

```
def display_courses(courses):  
    for course in courses:  
        print(course[0],  
              course[1],  
              course[4],  
              f"{course[5]:04d}",  
              course[2],  
              course[3])
```

The expression `f"{course[5]:04d}"` ensures that a time such as 930 is displayed as 0930.

4 Use the Module

In a second file or Google Colab notebook, begin with

```
import course_tools  
courses = []
```

Add the courses you are taking this semester, and the courses of a classmate. Use the format defined previously.

5 Sort the Schedule

Sort the courses by time:

```
course_tools.sort_by_time(courses)  
course_tools.display_courses(courses)
```

Predict which course will appear first and last.

6 Challenge

Extend the room-conflict function from the warm-up. Two courses should now be reported as a conflict only when they

- use the same building and room,
- meet at the same time, and
- share at least one meeting day.

For example, two courses in SLC 202 at 1400 should **not** conflict if one meets MWF and the other meets TR.